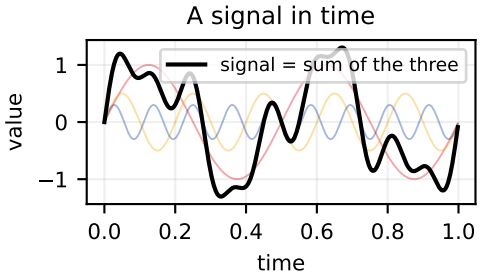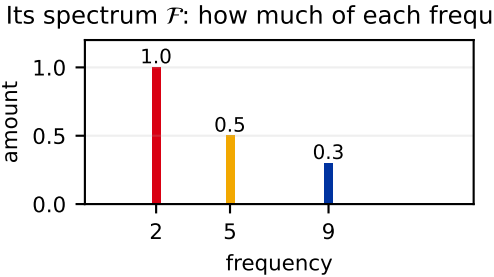The Fourier transform reads off the frequencies hiding in a signal